

1.0 COURSEWORK TITLE

Cyber intrusion detection and classification

THE COURSEWORK OVERVIEW

For the assignment, you are asked to explore the application of data analytics techniques to the dataset that is provided. You must study data problems related to the dataset, giving special consideration to the unique properties of the problem domain, and testing one or more techniques on it.

Your analysis needs to be thorough and go beyond the scope of what has been covered in this course. You should incorporate data exploration, manipulation, transformation, and visualization concepts with data analysis techniques in your solution. It is crucial to provide explanations and justifications for the chosen techniques.

You also may need to pre-process your data to get it into an appropriate format. The assignment should involve several techniques by categorizing it into different criteria and a detailed exploration of the commands used in each criterion. Outline the findings, analyze them, and justify them correctly with an appropriate graph. Also, a supporting document is needed to reflect the graph and code using R programming concepts.

2.0 OBJECTIVES OF THIS COURSEWORK

This assignment will help you to explore and analyze a set of data and reconstruct it into meaningful representations for decision-making.

3.0 TYPE

Group Assignment (4 members)

4.0 COURSEWORK DESCRIPTION

This dataset contains the hybrid of real modern normal activities and synthetic contemporary attack behaviours of cyber-attack. As a data analyst in the cybersecurity sector, you are commissioned to conduct an in-depth analysis of this dataset with 9 different types of attack

to identify the factors that differentiate the different type of attacks and determine the presence of attack.

Techniques

The dataset provided for this assignment consists of information about different set of factors such as flow features, content features, time features, connection features etc. Each features have its own different set of variables. Students are required to identify certain sets of variables to find the relationship between variable studied against the type of attacks or presence of attack. In addition to the techniques (data exploration, manipulation, transformation, and visualization techniques) covered in the course to analyze the dataset, you might consider exploring and implementing more advanced concepts to enhance the effectiveness of data analysis.

5.0 GENERAL REQUIREMENTS

- This is a group assignment, with a maximum of 4 students in a group.
- The R program should compile and be executed without errors.
- Validation should be done for each entry from the users to avoid logical errors.
- Do not use third-party tools such as Excel, OpenRefine and etc to pre-process or clean the data. Cleaning and pre-processing must be done in R using scripting.
- No duplication is allowed in the dataset.
- You should;
 - Include good programming practices such as comments, variable naming conventions, and indentation.
 - Carried out additional research from the Internet to comprehend the knowledge and information on the given dataset when examining the data.
- The analysis should be meaningful and effective in providing the information for decision-making.
- Any additional features implemented must improve the retrieval effects.

6.0 **DELIVERABLES:**

The complete **RScript** (source code) and **report** must be submitted to **the APU Learning Management System (Moodle)**.

6.1 **RScript (Program Code):**

- Name the file under your group number.
- Start the first few lines in your program by typing all member's names and TP numbers. For example:
Name1, TP000001
Name2, TP000002
Name3, TP000003
Name4, TP000004
- For each objective example, provide student id and explain what you want to discover. For example:
Objective 1: To investigate the relationship between the revision method and academic achievement.
Analysis 1-1: Which revision method are more effective?
Analysis 1-2: Is there a significant correlation between revision method and academic achievement?
Analysis 1-3: What are the external factor that interact with revision method to influence the academic achievement?
- For each extra feature example, give an id and provide the explanation.
Extra feature 1
comments about the extra feature

6.2 **Documentation (report):**

- ✓ A 6500 words (max 65 pages) report including appendix. The report should comprise of the following content.

Cover Page:

All reports must be prepared with a *front cover*. A protective transparent plastic sheet can be placed in front of the report to protect the front cover. The front cover should be presented with the following details:

Module

- ✍ Coursework Title
- ✍ Intake
- ✍ Students name and id
- ✍ Date Assigned (the date the report was handed out).
- ✍ Date Completed (the date the report is due to be handed in).

Contents:

- Introduction
 - ✓ Data Description
 - ✓ Assumptions (if any)
 - ✓ Hypothesis and Objectives
- Data Preparation
 - ✓ Data import
 - ✓ Cleaning / pre-processing (if necessary)
 - ✓ Data Validation (if necessary)
- Data Analysis
 - ✓ Each objective (along with student name) must start in a separate page and contains:
 - Analysis Techniques – e.g. descriptive using statistics
 - Screenshot of source code with output / plot.
 - Outline the findings based on the results obtained.
 - ✓ The extra feature explanation must be in a separate page and contains:
 - Screenshot of source code with output/plot.
 - Explain how adding this extra feature can improve the results.
 - ✓ Interpret the results from each analysis
- Conclusion
 - ✓ Overall discussion on the findings from all objectives
 - ✓ Recommendation
 - ✓ Limitation and future direction
 - ✓ State the word count (at the end of page)

*Workload Matrix**References*

- ✍ You may source algorithms and information from the Internet or books. Proper referencing of the resources should be evident in the document.

✎ All references must be made using the **APA (American Psychological Association)** referencing style as shown below:

- ✓ The font size used in the report must be 12pt and the font is Times New Roman.

7.0 **ASSIGNMENT ASSESSMENT CRITERIA**

The assignment assessment consists of 2 major components: Analysis (70%) and finding and discussion (30%). Details of the division for each component are as follows:

Analysis (70%)		Finding and Discussion (30%)	
Criteria	Marks Allocated	Criteria	Marks Allocated
<u>Methodology</u> ✓ Approaches used to process, interpret, and extract insights from data. <u>Report content:</u> → Methodology description	10%	<u>Finding and Discussion</u> ✓ Project Introduction (description, assumption, hypothesis, objectives) ✓ Conclusion (result finding, discussion, recommendation and future direction) ✓ Structure of the report and references	30%
<u>Analysis Methods</u> → Specific process employed to carry out the analysis (transform raw data into meaningful insights) <i>[e.g. Exploratory data analysis, Hypothesis test, Descriptive statistics etc]</i> <u>Report content:</u> → RScript – code snippet and visualization screenshots with explanation.	60%		

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Adjust the rest of the marks.
 Program solution : 40%
 Presentation : 20%

8.0 **DEVELOPMENT TOOLS**

The program written for this assignment should be written in R Studio

9.0 **ACADEMIC INTEGRITY**

- You are expected to maintain the utmost level of academic integrity during the duration of the course.
- **Plagiarism is a serious offence and will be dealt with according to APU and De Montfort University regulations on plagiarism.**